Dataset and setup: SST-2

SST-2: binary sentiment

67,349 train rows: 6,885 complete sentences with their sub-phrases (77 % share the sentence's label)

872 validation sentences (49 % negative, 51 % positive -> accuracy is a valid metric)

1,821 test sentences, labels hidden: not usable

- Train: 44 % negative, 56 % positive; majority baseline on validation 0.509
- Word vocabulary 14,801 types; 5.7 % of validation tokens unseen
- 371 duplicate rows, 5 conflicting labels; no overlap with validation

Configuration

Choice	Setting
Tokenizers	NLTK words; BPE (trained on SST-2) with 500, 2k, 8k and 20k tokens
Features	token indicators, bag of words, mean GloVe
Models	log-linear, MLP, linear SVM, tuned MLP, 2-layer MLP, tuned C SVM
MLP	32 units, dropout 0.3, AdamW (lr 1e-4, wd 1e-2), 10 epochs
SVM	linear, $C = 1$
Evaluation	5-fold CV on whole-sentence folds, 3 seed(s) for NLTK systems (1 for BPE); accuracy
Analysis	Bayesian correlated t-test, ROPE of 0.005

Tokenization

Vocabulary and coverage

Tokenizer	Vocabulary	Tokens/sentence	Unknown
word (NLTK)	14,801	19.6	5.70 %
BPE 500	500	42.1	0.01 %
BPE 2000	2,000	30.3	0.01 %
BPE 8000	8,000	23.7	0.01 %
BPE 20000	20,000	21.6	0.02 %

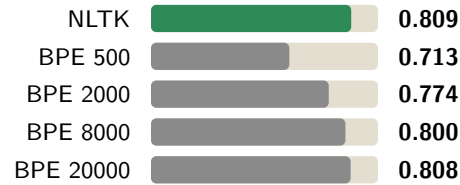
Validation sentences. BPE trained on SST-2 train, min. merge frequency 2.

Bayesian comparison ROPE = 0.005

vs NLTK	NLTK wins	Tie	BPE wins
BPE 500	1.00	0.00	0.00
BPE 2000	0.99	0.01	0.01
BPE 8000	0.96	0.04	0.00
BPE 20000	0.14	0.80	0.06

Accuracy by tokenizer

Log-linear, token indicators, CV on parent folds. Bars start at 0.50.



- BPE 20000 is almost the same as NLTK
- BPE 500, 2000 and 8000 are worse than NLTK
- The tutorial's AG News tokenizer gives 26 % more chunks on SST-2 validation than an SST-2 BPE of the same size
- We use NLTK

Representations: indicators, counts and GloVe

Token indicators

1 if the token occurs in the sentence, otherwise 0

Dimension: vocabulary (14,802 for NLTK)

Sparse

0.809

CV accuracy, log-linear, 3 seeds

Bag of words

How often each token occurs in the sentence

Dimension: vocabulary (14,802 for NLTK)

Sparse

0.808

CV accuracy, log-linear, 3 seeds

Mean GloVe

Average of the 300-d GloVe vectors of the words (our own choice)

Dimension 300, standardised

Dense, NLTK tokens only

0.787

CV accuracy, log-linear, 3 seeds

- Indicators and bag of words are practically equivalent (P(tie) = 0.86) (only 7.3 % of features have a count above 1)
- Mean GloVe is worse than indicators (P = 0.99)
- GloVe averages over word GloVe vectors -> word order and therefore sentiment is lost.
- Considered but not tried: TF-IDF, n-grams, a learned embedding layer

Architectures: log-linear, MLP and linear SVM

Mean CV accuracy (NLTK)

Model (NLTK)	Indic.	BoW	GloVe
Log-linear	0.809	0.808	0.787
MLP	0.809	0.808	0.790
Linear SVM	0.790	0.789	0.786
MLP, tuned	0.804	0.803	0.790
2-layer MLP	0.806	0.803	0.794
SVM, tuned C	0.801	0.789	0.787

Tuned MLP: lr 1e-3, 128 units, early stopping. Tuned C: 0.3 (indicators), 1 (BoW), 0.01 (GloVe).

- Evaluated dropout [0, 0.1, 0.5, 0.7] on bag of words: 0: 0.802 0.7: 0.811, a trend (P = 0.77) but no certainty.
- Activation on MLP BoW: (tanh, sigmoid, GELU) makes no difference. With identity activation the MLP loses only 0.5 points

Bayesian comparison (CV), ROPE 0.005

Log-linear vs	Log-lin. wins	Tie	Other wins
MLP, indicators	0.20	0.61	0.20
MLP, bag of words	0.11	0.78	0.11
MLP, mean GloVe	0.02	0.70	0.28
MLP, tuned, indicators	0.53	0.42	0.04
MLP, tuned, bag of words	0.50	0.46	0.04
MLP, tuned, mean GloVe	0.05	0.57	0.38
2-layer MLP, indicators	0.36	0.60	0.04
2-layer MLP, bag of words	0.48	0.47	0.05
2-layer MLP, mean GloVe	0.03	0.35	0.62
SVM (C = 1), indicators	1.00	0.00	0.00
SVM (C = 1), bag of words	1.00	0.00	0.00
SVM (C = 1), mean GloVe	0.02	0.98	0.01
SVM, tuned C, indicators	0.87	0.13	0.00
SVM, tuned C, bag of words	1.00	0.00	0.00
SVM, tuned C, mean GloVe	0.01	0.99	0.01

- For hidden layer vs. log-lin most of posterior lies inside ROPE, so no evidence of improvement
- For linear SVM(C = 1) vs. log-linear, log-linear is better on indicators and bag of words. Mean GloVe is a tie.
- Tuning the MLP (learning rate, size) or adding a second layer gives no reliable gain
- Tuning C on indicators -> the SVM improves from 0.790 to 0.801, P = 0.94. For bag of words -> the best C = default C=1.

Evaluation: cross-validation without leakage

Parent folds

- Find the 6,885 parent sentences: rows that are not inside a longer row
- Split the parents into folds; each held-out fold is whole
- Train on all other rows, minus held-out sentences and their sub-phrases
- MLP: epoch chosen on an inner 10 % slice, never on the held-out fold

Random folds leak

Log-linear, NLTK, indicators, seed 0. Bars start at 0.50.



- Random folds give higher CV accuracy through leakage
- Metric: accuracy because of balanced classes
- 3 seed(s) $\times$ 5 folds $=$ 15 paired scores for chosen NLTK
- Tokenizer built on all of train: no inflation (per-fold vocabulary changes $\leq$ 0.003)
- Validation: 872 sentences (SE $\approx$ 0.013)

Analysis and choice

Pairwise results

Comparison (A vs B)	A wins	Tie	B wins
NLTK vs BPE 20k, log-lin. (1 seed)	0.136	0.800	0.064
NLTK vs BPE 2k, log-lin. (1 seed)	0.985	0.009	0.006
NLTK Indicators vs BoW, log-lin.	0.116	0.857	0.027
NLTK Indicators vs GloVe, log-lin.	0.986	0.013	0.001
Log-lin. vs MLP, BoW	0.113	0.782	0.105
Log-lin. vs 2-layer MLP, indic.	0.362	0.597	0.041
Log-lin. vs SVM C=1, indic.	0.998	0.002	0.000
Log-lin. vs SVM tuned, indic.	0.870	0.130	0.000

- ROPE(0.005) $\approx$ 7 held-out sentences, below the fold noise (sd $\approx$ 0.01)
- No clear winner for log-linear and MLPs on indicators/counts (CV 0.803 to 0.809)
- Log-linear = MLP is undecided at ROPE 0.005 but practically equivalent at 0.01

Choice

- **Log-linear with token indicators on NLTK tokens**: highest CV accuracy (0.809), tied with the MLPs, simplest, fastest
- Validation (872 sentences, SE $\approx$ 1.3 points): MLPs highest, but it only one split

Other aspects of performance

- **Class bias:** 5 of 6 systems predict 53–55 % positive (validation: 50.9 %)
- **Hard sentences:** negation -12 points, long sentences -9 to -15, contrast about -8. Unknown words: no effect
- **Shared mistakes:** 71 sentences are wrong for every system, the representation decides which sentences go wrong, not the architecture
- **Weights, time:** *not* -> fixed negative weight, *half-bad* -> positive, *but* -> weight 0.
- **MLPs** are 3-7 times slower, for no gain